Pre-Analysis Plan

Research Team: Thorsten Kneip and Gerrit Bauer (Team ID:

– No data Group –

ANNOTATED VERSION

When preparing the pre-analysis plan, the code was written without access to the SOEP data. After obtaining the 5-percent subsample, we identified several minor discrepancies between the anticipated data structure and the actual SOEP variables, as well as some typographic errors in the original PAP. As a result, the originally proposed Stata code did not run without errors. Below, we document all adjustments that were necessary to implement the PAP as closely as possible in the available data. These notes clarify which issues arose, why they occurred, and how they were resolved.

Specifically, we complement the preliminarily proposed code with the final code, including how the analytic sample is selected.

Noteworthy, there are no substantive deviations from the original PAP. The largest discrepancy refers to the necessity of imposing parametric constraints in testing H3 & H4, which was already anticipated in the PAP. We aimed at greatest flexibility to avoid issues of functional form misspecification but did not attempt more refined methods to determine the functional form in the small subsample at hand. We briefly considered switching to matching or IPW but abandoned this in order to remain fully compliant with the PAP.

Added comments to the PAP are marked in yellow.

Subjects to changes in the original PAP are marked in turquoise.

Section I: Theoretical estimands, identification, and (refined) empirical estimands

THEORETICAL ESTIMANDS

We distinguish two types of estimands: (I) population average treatments effects of having a daughter (vs. not having a daughter) and (II) gender specific average treatment effects (and their differences, respectively).

Theoretical estimand I:

$$ATE(D) = \frac{1}{n}\sum_{i=1}^{n}(Y_i^1 - Y_i^0),$$

where $D = \begin{cases} 0 & having\ no\ daughters \\ 1 & having\ at\ least\ one\ daughter \end{cases}$

and $Y = \begin{cases} attitude\ promoting\ gender\ equality \\ policy\ preference \\ bahavior\ promoting\ gender\ equality \end{cases}$

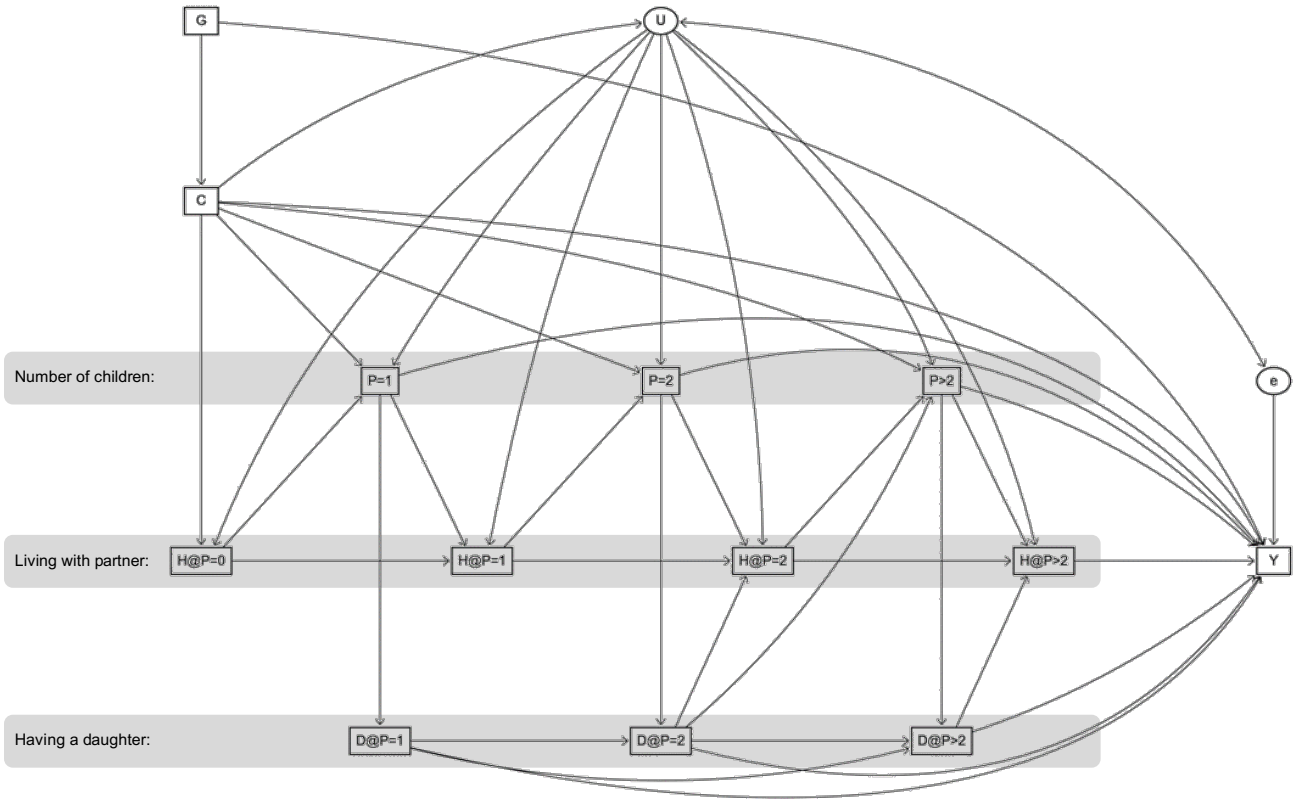
Theoretical estimand II:

$$ATE_{\Delta G}(D) = \frac{1}{n}\sum_{i=1}^{n}\left((Y_i^{11} - Y_i^{01}) - (Y_i^{10} - Y_i^{00})\right),$$

where $G = \begin{cases} 0 & male \\ 1 & female \end{cases}$

IDENTIFICATION

We assume the following data generating process:



Key features/assumptions:

- Number of children ("parity") is an endogenous choice variable; it is non-causally related to the outcomes of interest via unobservables U (e.g., family orientation).
- Child gender is randomly assigned at any given parity.
- The gender of the first child does neither influence the decision to have a 2nd child nor the probability of cohabitation
  [*exclusion restriction set 1* : D@P=1 ⊥ P2; D@P=1 ⊥ H@P=1 | P=1],
- … but may do so at higher parities.
- Men and women do not differ in their number of children, living arrangement, or unobservable characteristics given observable characteristics (C, e.g., education)
  [*exclusion restriction set 2* : G ⊥ P | C; G ⊥ U | C; G ⊥ H | C].

Exclusion restriction set 1 is testable.
Exclusion restriction set 2 is partially testable (i.e., w.r.t. number of children and living arrangement).

In principle, this can be utilized to inform adjustment variable selection (C).

Under this DGP, the average causal effect of having a daughter is *not identified* in the total population. However, under exclusion restriction 1, it is identified in the population of parents of a single child. This assumption seems reasonable for Germany, given the empirical evidence on little to no child gender preferences but a strong preference for a mixed-sex composition.[1]

Empirical Estimands

The respective empirical estimand is then given by:

$$E[ATE(D)|\, P = 1] = E[Y|D = 1, P = 1] - E[Y|D = 0, P = 1]$$

The ATE of having a daughter among parents of one child will equal the ATE in the total population in the absence of effect heterogeneity by parity; it will equal the ATE among the parent population if effects differ between parents and non-parents but are homogenous among parents.

To identify gender specific effects in the single child population, we have to additionally impose exclusion restriction set 2. If these do not hold, identification will fail due to collider stratification bias. We propose to control for education and age/cohort

If they hold, the respective empirical estimand for gender differences in effects of daughters is given by

$$\begin{aligned} E[ATE_{\Delta G}(D) \mid P = 1, c] \\ &= (E[Y|D = 1, G = 1, P = 1, c] - E[Y|D = 1, G = 1, P = 1, c]) \\ &- (E[Y|D = 1, G = 0, P = 1, c] - E[Y|D = 1, G = 0, P = 1, c]) \end{aligned}$$

[1] We see these assumptions largely in line with the findings reported in Dahl and Moretti (2008). Based on US census data (N ≈ 4 Million) they find no effect of the first child being a daughter on having two or more children. There is some evidence on effects on household composition (living with fathers), but effects are very small and diminishing over cohorts. If also present in our data, we would thus only expect very small potential bias.

Section II: Data

Variable Name	Variable Type	Original Variable(s)	Dataset	Hypothesis	Year(s)	Description of Preparation
daughters	treatment	kidsex01	biobirth	H1-H4	all	daughters=1 if intdate-birthdate_1>0 & kidsex01==2 daughters=0 if intdate-birthdate_1>0 & kidsex01==1 daughters=. if intdate-birthdate_1<=0 daughters=. if intdate>=. daughters=. if birthdate_01>0. daughters=. if birthdate_01>=. /*typo-correction*/ (see help variables below)
famvscareer	outcome	plh0301	pl	H1, H4	2012	recode plh0301 (-8 -5 -1=.) (4=1) (3=2) (2=3) (1=4), gen (famvscareer) high values indicate attitudes towards gender equality No recoding was necessary, as higher values in the original variable already indicated greater gender equality: recode plh0301 (-100/-1=.), gen (famvscareer)
menhousework	outcome	plh0303	pl	H1, H4	2012	recode plh0303 (-8 -5 -1=.) (4=1) (3=2) (2=3) (1=4), gen(menhousework) high values indicate attitudes towards gender equality
left	outcome	plh0004	pl	H2	2005, 2009, 2014, 2019, 2020, 2023	recode plh0004 (-8 -5 -1=.) (10=0) (9=1) (8=2) (7=3) (6=4) (5=5) (4=6) (3=7) (2=8) (1=9) (0=10), gen(left) high values indicate left-leaning political preferences
workhours	outcome	pgtatzeit, pgemplst	pgen	H3	all	Negative values in variable pgtatzeit coded to missing. Working time set to 0 if pgemplst==5 (not working)
housework	outcome	pli0043_h	pl	H3	all	Negative values coded to missing. Daily housework hours top coded to a maximum of 16 hours.
cohab	sample	pgpartz	pgen	H3	all	cohab=1 if pgpartz!=0 & pgpartz!=5 restrict analysis to cohabiting respondens
parity2	sample			H1-H4	all	restrict analysis to respondents with one biological child: parity2=1 if intdate-birthdate_2>0 (see help variables below)
birthyear	control	gebjahr	ppath	H2, H3	all	cleaning of missing values gebjahr was retrieved from ppathl instead of ppath as planned in the PAP (identical information)
educ	control	pgisced97	pgen	H1-H4	all	recode into 3 categories: low: 0, 1, 2 middle: 3, 4 high: 5, 6
female	control (H1-H3) and interaction (H4)	sex	ppathl	H1-H4	all	female=1 if sex==2 female=0 if sex==1 female=. If sex<=0

Variable Name	Variable Type	Original Variable(s)	Dataset	Hypothesis	Year(s)	Description of Preparation
phrf	weight	phrf	ppathl	H1-H4	all	original variable used as is, weighting on individual level
pid	identifier	pid	all		all	person identifier, used for merging
cid	identifier	cid	all	H1-H4	all	household of origin identifier, used for clustered standard errors
syear	identifier/control	syear	all	H2, H3		used for merging, control when pooled data are analysed
birthdate_1	help	kidgeb01, kidmon01	biobirth		all	birthdate of first for first child, day set to 15th of month, month set to 6 if missing
birthdate_2	help	kidgeb02, kidmon02	biobirth		all	same procedure for second child
intdate	help	pgpiyear, pgmonth	pgen		all	date of interview, day set to 15th of month We detected a high number of missing interview month values only after accessing the data and imputed February, the modal month, for all missing cases. replace pgmonth=2 if pgmonth>=.

Section III: Estimation and hypothesis testing

(H1)

Estimation

We estimate the following regressions in the sample of respondents with a single child (`parity2=1`). For appropriate inference, we apply weights cross-sectionally calibrated weights (`phrf`). To account for non-independence of observations, we cluster standard errors at the household level:

(a) $famvscareer = \alpha_{1a} + \beta_{1a} daughters + \gamma'_{1a} C + \delta'_{1a} daughters \times C + \varepsilon_{1a}$

(b) $menhousework = \alpha_{1b} + \beta_{1b} daughters + \gamma'_{1b} C + \delta'_{1b} daughters \times C + \varepsilon_{1b}$

where C is a vector of control variables centered at their sample means.

Under the assumptions outlined above, no further adjustment is necessary, but controlling for additional variables (i.e., `female`, `educ`, and `birthyear`) would be possible and could add precision to estimates.

Given that `famvscareer` and `menhousework` are measured on a 4-point scale, we deem ordered regression appropriate.

Hypothesis testing

The corresponding Null to test against is $\{\beta_{1a}, \beta_{1b}\} \leq 0$.

I.e., to evaluate (H1), we need to derive the test statistics for one-sided tests of β_{1a} and β_{1b}, adjusted for multiple testing. We apply the method proposed by Drukker e al. (2024). We would reject the null if any of β_{1a}, β_{1b} would fall outside the joint confidence band (at a given a priori a confidence level; α=.05).

Preliminary Stata code

```
oprobit famvscareer daughters if parity2==1 [pw=phrf]
est sto e1

oprobit menhousework daughters if parity2==1 [pw=phrf]
est sto e2

suest e1 e2, vce(cluster cid)

margins, dydx(daughters) predict(xb equation(e1_famvscareer)) ///
predict(xb equation(e2_menhousework)) post

sotable, alternative(upper)
```

Final Stata code

```
*sample restrictions:
keep if parity2!=1
 // only respondents with a single child
keep if !missing(daughters, phrf) & phrf!=0
 // only respondents with valid information on treatment status
    and sampling weight
keep if !missing(famvscareer) | !missing(menhousework)
 // only respondents with non-missing outcome information

*analysis
oprobit famvscareer daughters [iw=phrf]
est sto e1

oprobit menhousework daughters [iw=phrf]
est sto e2

suest e1 e2, vce(cluster cid)

margins, dydx(daughters) predict(xb equation(e1_famvscareer)) ///
predict(xb equation(e2_menhousework)) post

sotable, alternative(upper)
```

Deviations from preliminary code:

- Variable parity2 takes value 1 if 2nd child is born, i.e., the corresponding sample restriction translates to `parity2!=1`; this was a mistake in the original PAP.
- `suest` takes iweights rather than pweights.

(H2)

Estimation

We estimate the following regression in the sample of respondents with a single child (`parity2=1`). For appropriate inference, we apply weights cross-sectionally calibrated weights (`phrf`). To account for non-independence of observations, we cluster standard errors at the household level:

$left = \alpha_2 + \beta_2 daughters + \gamma_2' C + \delta_2' daughters \times C + \varepsilon_2$

where C is a vector of control variables centered at their sample means.

Under the assumptions outlined above, no further adjustment is necessary, but controlling for additional variables (i.e., `female`, `educ`, and `birthyear`) would be possible and could add precision to estimates. We propose to pool and average across waves with available information on the outcome (thus C includes `syear`).

Given that `left` is measured on an 11-point scale, we deem linear regression appropriate.

Hypothesis testing

The corresponding Null to test against is $\beta_2 \leq 0$, which is again one-sided.

Preliminary Stata code

```
regress left daughters##syear if parity2==1 [pw=phrf], vce(cluster cid)

margins, dydx(daughters) at((mean) syear) post

sotable, alternative(upper)
```

Final Stata code

```
*sample restrictions:
keep if parity2!=1
 // only respondents with a single child
keep if !missing(left, daughters, syear, phrf) & phrf!=0
// only respondents with non-missing information on outcome, treatment status,
   and sampling weight

*analysis
regress left daughters##syear [pw=phrf], vce(cluster cid)

margins, dydx(daughters) at((mean) syear) post

sotable, alternative(upper)
```

Deviations from preliminary code:

- Variable parity2 takes value 1 if 2nd child is born, i.e., the corresponding sample restriction translates to `parity2!=1`; this was a mistake in the original PAP.

(H3)

Estimation

We estimate the following regressions in the sample of respondents with a single child (`parity2=1`) and living with a partner (`cohab=1`). We impose the latter sample restrictions to assure comparability of outcome measures. For appropriate inference, we apply weights cross-sectionally calibrated weights (`phrf`). To account for non-independence of observations, we cluster standard errors at the household level:

(a) $$workhours = \alpha_{3a} + \beta_{3a} daughters + \gamma_{3a} female + \delta_{3a} daughters \times female + \zeta'_{3a} C + \eta'_{3a} daughters \times C + \vartheta'_{3a} daughters \times female \times C + \varepsilon_{3a}$$

(b) $$housework = \alpha_{3b} + \beta_{3b} daughters + \gamma_{3b} female + \delta_{3b} daughters \times female + \zeta'_{3b} C + \eta'_{3b} daughters \times C + \vartheta'_{3b} daughters \times female \times C + \varepsilon_{3b}$$

where C is a vector of control variables centered at their sample means.

We propose to include control variables `female`, `educ`, and `birthyear` as additional controls to lend more credibility to the CIA w.r.t. `female` and again pool and average over available waves

Given that `workhours` and `housework` are measured in hours, we deem linear regression appropriate.

Hypothesis testing

The corresponding Null to test against is:

$\beta_{3a} \geq 0$	[Fathers of daughters do NOT work less]
$(\beta_{3a} + \delta_{3a}) \leq 0$	[Mothers of daughters do NOT work more]
$\beta_{3b} \leq 0$	[Fathers of daughters do NOT do more housework]
$(\beta_{3b} + \delta_{3b}) \geq 0$	[Mothers of daughters do NOT do less housework]

I.e., to evaluate (H3), we need to derive the test statistics for one-sided tests of $-\beta_{3a}$, $(\beta_{3a} + \delta_{3a})$, β_{3b}, and $-(\beta_{3a} + \delta_{3a})$, adjusted for multiple testing, see (H1).

Preliminary Stata code

```
regress workhours daughters##female##syear##educ##c.birthyear ///
if parity2==1 & cohab==1 [pw=phrf]
est sto e1

regress housework daughters##female##syear##educ##c.birthyear ///
if parity2==1 & cohab==1 [pw=phrf]
est sto e2

suest e1 e2, vce(cluster cid)

margins, dydx(daughters) predict(equation(e1_mean)) ///
predict(equation(e2_mean)) at(female=(0 1) (mean) syear educ birthyear) post

nlcom (-_b[1.daughters:1._predict#1._at]) (_b[1.daughters:1._predict#2._at]) ///
(_b[1.daughters:2._predict#1._at]) (-_b[1.daughters:2._predict#2._at]), post

sotable, alternative(upper)
```

Final Stata code

```
*sample restrictions:
keep if parity2!=1
 // only respondents with a single child
keep if cohab==1
 // only cohabiting respondents
keep if !missing(daughters, female, syear, educ, birthyear, phrf) & phrf!=0
// only respondents with non-missing information on outcome, treatment status,
   control variables, and sampling weight

*analysis
regress workhours (daughters##female)##(syear educ c.birthyear) [iw=phrf]
est sto e1

regress housework (daughters##female)##(syear educ c.birthyear) [iw=phrf]
est sto e2

suest e1 e2, vce(cluster cid)

margins, dydx(daughters) predict(equation(e1_mean)) ///
predict(equation(e2_mean)) at(female=(0 1) (mean) syear educ birthyear) post

nlcom (-_b[1.daughters:1._predict#1._at]) (_b[1.daughters:1._predict#2._at]) ///
(_b[1.daughters:2._predict#1._at]) (-_b[1.daughters:2._predict#2._at]), post

sotable, alternative(upper)
```

Deviations from preliminary code:

- Variable parity2 takes value 1 if 2nd child is born, i.e., the corresponding sample restriction translates to `parity2!=1`; this was a mistake in the original PAP.
- `suest` takes iweights rather than pweights.
- There is one deviation linked to an already anticipated risk (see below). Because full stratification was not feasible in the 5% sample, we constrained the model to only include the daughters × female interaction and all pairwise and triple interactions of these with the additional covariates.

(H4)

Estimation

We estimate the following regressions in the sample of respondents with a single child (`parity2=1`). For appropriate inference, we apply weights cross-sectionally calibrated weights (`phrf`). To account for non-independence of observations, we cluster standard errors at the household level:

(a) $$famvscareer = \alpha_{4a} + \beta_{4a} daughters + \gamma_{4a} female + \delta_{4a} daughters \times female + \zeta'_{4a} C + \eta'_{4a} daughters \times C + \vartheta'_{4a} daughters \times female \times C + \varepsilon_{4a}$$

(b) $$menhousework = \alpha_{4b} + \beta_{4b} daughters + \gamma_{4b} female + \delta_{4b} daughters \times female + \zeta'_{4b} C + \eta'_{4b} daughters \times C + \vartheta'_{4b} daughters \times female \times C + \varepsilon_{4b}$$

where C is a vector of control variables centered at their sample means.

We propose to include control variables `female`, `educ`, and `birthyear` as additional controls to lend more credibility to the CIA w.r.t. `female`.

Hypothesis testing

The corresponding Null to test against is $\{\delta_{4a}, \delta_{4b}\} \geq 0$

I.e., to evaluate (H4), we need to derive the test statistics for one-sided tests of δ_{4a} and δ_{4b}, adjusted for multiple testing, see (H1).

Preliminary Stata code

```
oprobit famvscareer daughters##female##educ##c.birthyear if parity2==1 [pw=phrf]
est sto e1

oprobit menhousework daughters##female##educ##c.birthyear if parity2==1 [pw=phrf]
est sto e2

suest e1 e2, vce(cluster cid)

margins, dydx(daughters) predict(xb equation(e1_famvscareer)) ///
predict(xb equation(e2_menhousework)) at(female=(0 1) (mean) educ birthyear) post

sotable, alternative(lower)
```

Final Stata code

```
*sample restrictions:
keep if parity2!=1
 // only respondents with a single child
keep if !missing(daughters, educ, birthyear, phrf) & phrf!=0
// only respondents with non-missing information on outcome, treatment status,
   control variables, and sampling weight
keep if !missing(famvscareer) | !missing(menhousework)
 // only respondents with non-missing outcome information

*analysis
oprobit famvscareer i.daughters i.female i.educ c.birthyear daughters#female ///
daughters#educ daughters#c.birthyear female#educ female#c.birthyear ///
educ#c.birthyear [iw=phrf]
est sto e1

oprobit menhousework i.daughters i.female i.educ c.birthyear daughters#female ///
daughters#educ daughters#c.birthyear female#educ female#c.birthyear ///
educ#c.birthyear [iw=phrf]
est sto e2

suest e1 e2, vce(cluster cid)

margins, dydx(daughters) predict(xb equation(e1_famvscareer)) ///
predict(xb equation(e2_menhousework)) at(female=(0 1) (mean) educ birthyear) post

nlcom (_b[1.daughters:1._predict#1._at] - _b[1.daughters:1._predict#2._at]) ///
(_b[1.daughters:2._predict#1._at] - _b[1.daughters:2._predict#2._at]), post

sotable, alternative(upper)
```

Deviations from preliminary code:

- Variable parity2 takes value 1 if 2nd child is born, i.e., the corresponding sample restriction translates to `parity2!=1`; this was a mistake in the original PAP.
- `suest` takes iweights rather than pweights.
- As with H3, there is one deviation linked to an already anticipated risk (see below). Because full stratification was not feasible in the 5% sample, we constrained the model to only include all pairwise interactions (in addition to all main effects) in the model and omit higher order interactions.
  We did not attempt more refined methods to determine the functional form in the small subsample at hand. We briefly considered switching to matching or IPW but abandoned this in order to remain fully compliant with the PAP.
- Furthermore, the preliminary code in the PAP mistakenly did not include the contrasting of effects by gender yielding the effects of interest.

Note

According to our reading of the task description, the above proposal neglects that a joint assessment of H1-H4 in non-independent samples would require an additional adjustment for multiple testing.

Perceived risks

We identify two major challenges in pursuing the above proposal:

(1) The outcome variables proposed to asses (H1) and (H4) exhibit a large fraction of missing values. This may lead to too small sample sizes, particularly in the evaluation of (H4) and given that our identification strategy requires an additional sample restriction. Moreover, and even if sample sizes turn out to be sufficient, we could not trace the applied routing and hence could not evaluate whether the resulting sample selection may create bias.

(2) The regression models proposed above are only minimally constrained. In practice, this may create empty cells and require imposing constraints (like the omission of interaction terms).

(3) The testable assumptions implied by our proposed DGP may not hold in the data. However, we cannot ultimately asses this based on a (5%) subsample; neither can we refine our control set to assure assumptions hold.